

Peter W. Fields

Physicist | Educator

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SUMMARY

- Two years teaching experience in introductory physics for undergraduates, including leading lab sections, discussion sections, and exam reviews.
- PhD researcher in statistical physics / machine learning for applications in biology.
- Cross-disciplinary research and science communication across physics, machine learning, biology, and neuroscience — including presentations at international conferences.

EDUCATION

PhD in Physics, University of Chicago – March 2026

Bachelor of Science in Physics, The City College of New York – 2019

TEACHING EXPERIENCE

Teaching Assistant, General Physics I: Mechanics – Autumn 2019, Autumn 2021

Teaching Assistant, General Physics II: Electricity and Magnetism – Winter 2019, 2020 & 2021

Led weekly discussion and lab sections, held office hours, gave exam review sessions, and graded problem sets and exams.

OCIA Instructor at Calvert House – Fall 2024 through Easter 2025

Led lessons on selected topics from the Catechism. Worked on a team of instructors and helped answer questions during other instructors' lessons. Met weekly throughout the Liturgical year.

RESEARCH EXPERIENCE

Palmer Theory Group, University of Chicago (Jan 2021 – Present)

- Applied statistical physics and machine learning methods to problems in protein modeling, computational neuroscience, and quantitative biology
- Developed synthetic benchmarks and probabilistic models to study generalization and emergent structure in biological sequence data
- Cross-disciplinary collaboration with biologists, neuroscientists, and AI researchers

Messinger Research Group, The City College of New York (Jan 2018 – Aug 2019)

- Physics-based characterization of novel battery technologies
- Designed and executed impedance spectroscopy experiments
- Studied diffusive, kinetic, and electrical properties of battery systems

SELECTED TALKS

Temperature-tuning trained energy functions improves generative performance

APS Global Physics Summit 2025 – **P. W. Fields**, V. Ngampruetikorn, D. J. Schwab, S. E. Palmer

Characterizing Inference of Non-reciprocal Connections in the Kinetic Ising Model

APS March Meeting 2024 – **P. W. Fields**, Cheyne Weis, S. E. Palmer, Peter Littlewood

PUBLICATIONS & PREPRINTS

Understanding temperature tuning in energy-based models

P. W. Fields, V. Ngampruetikorn, D. J. Schwab, S. E. Palmer.

Preprint (2025). arXiv:2512.09152

Code: github.com/peter-fields/temp-tune

Understanding Energy-Based Modeling of Proteins via an Empirically Motivated Minimal Ground Truth Model

P. W. Fields, V. Ngampruetikorn, R. Ranganathan, D. J. Schwab, S. E. Palmer.

Synergy of Scientific and Machine Learning Modeling Workshop at International Conference in Machine Learning (2023). openreview.net/forum?id=vxn5QGPFyi

Code: github.com/peter-fields/toysector

Tunable Pseudocapacitive Intercalation of Chloroaluminate Anions into Graphite Electrodes for Rechargeable Aluminum Batteries

J. H. Xu, T. Schoetz, J. R. McManus, V. R. Subramanian, **P. W. Fields**, R. J. Messinger.

Journal of the Electrochemical Society 168, 060514 (2021)

SKILLS

Graduate Coursework: Quantum Field Theory, General Relativity, Advanced Statistical Mechanics, Biophysics, Computational Neuroscience

Coding: Python, Julia, Mathematica, LaTeX

AWARDS

Soft and Living Matter Exchange Award, Institute for Complex Adaptive Matter (Dec 2023)

Funded 3-week exchange to work with a collaborator, David J. Schwab

Center for Physics of Evolving Systems Fellowship, University of Chicago (Jan 2022–Dec 2023)

For investigators at the interface of physics and biology, focusing on principles of the evolution of biological function.